

INTRODUCTION TO STATISTICAL METHODS · COMPANION READER

Probability Distributions

A field guide to the shapes of randomness: Bernoulli, Binomial, Poisson, Normal — and a first hello to χ^2 , t and F

Session 6 (30th–31st May 2026) · Read alongside the lecture slides · Every slide example worked out in full

How to use this companion. The slides give you the formulas; this reader gives you the *why*. Each distribution is introduced with an everyday picture first, then the math, then every example from the slides solved slowly enough that you can follow each step. Blue boxes carry the core idea, amber boxes carry analogies, green boxes carry worked examples, red boxes flag classic mistakes. If you remember only the purple boxes, you will still walk into the exam in good shape.

1 The big idea: distributions are reusable moulds

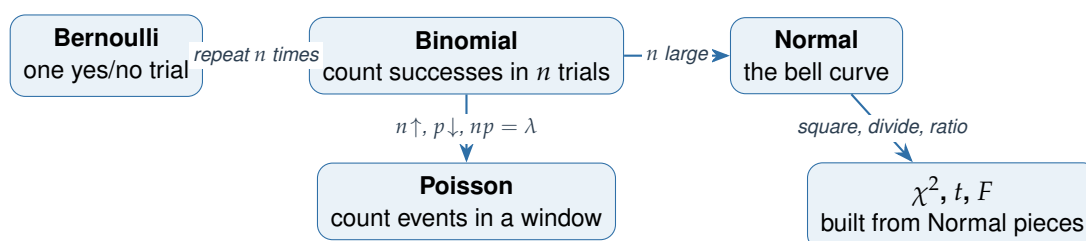
Randomness feels lawless, but it repeats itself. The number of heads in 10 coin tosses, the number of defective phones in a batch of 10, and the number of patients out of 10 who respond to a drug all behave *identically* — not because coins, phones and patients are alike, but because the underlying situation is the same: *a fixed number of independent yes/no trials with a constant success chance*.

A probability distribution is a **mould**: a reusable template that tells you, for every possible outcome, how likely it is. Once you recognise which mould your situation fits, all the work is already done for you — the formula, the mean, the variance are sitting there waiting. Learning this session well is mostly learning to *recognise moulds*.

★ Everyday picture

Think of distributions like dress patterns in a tailor's shop. The tailor does not redesign a shirt from scratch for every customer — he recognises “this is a shirt situation” and pulls out the shirt pattern, adjusting only the measurements. Here, the measurements are called **parameters**: p for Bernoulli, (n, p) for Binomial, λ for Poisson, (μ, σ^2) for Normal. Same pattern, different sizes.

This session's cast of characters, and how they are related:



One distinction runs through everything: **discrete** distributions (Bernoulli, Binomial, Poisson) count things — outcomes you could list: 0, 1, 2, ... — while **continuous** distributions (Normal, t , χ^2 , F) measure things — heights, times, weights — where between any two values lie infinitely many more.

2 Bernoulli: the atom of randomness

The intuition

A Bernoulli trial is the smallest possible random experiment: *one* attempt, *two* outcomes. A light switch is either on or off. A penalty kick scores or misses. An email gets opened or ignored. We label success 1 (probability p) and failure 0 (probability $q = 1 - p$). That is the entire distribution — two bars.

Formally, X has a Bernoulli distribution if its probability mass function (PMF) is

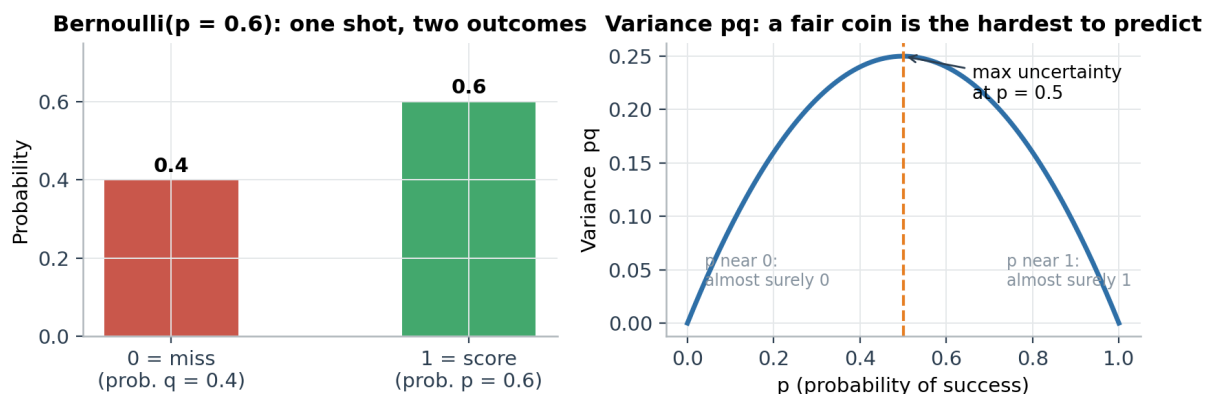
$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

The formula looks fancier than it is — it is just a compact way to write two facts at once. Plug in $x = 1$: you get $p^1(1 - p)^0 = p$. Plug in $x = 0$: you get $p^0(1 - p)^1 = 1 - p$. The exponents act like a switch that turns the right factor on.

Mean and variance, and what they are telling you

$$E[X] = 0 \cdot q + 1 \cdot p = p, \quad \text{Var}(X) = E[X^2] - E[X]^2 = p - p^2 = pq.$$

The mean is p for a transparent reason: if a striker scores 60% of penalties, then “on average” each kick is worth 0.6 goals. The variance pq is more interesting — it is a built-in *unpredictability meter*.



★ Everyday picture

Variance pq peaks at $p = 0.5$. A coin that lands heads 99% of the time is boring — you basically know what will happen, so variance is tiny ($0.99 \times 0.01 \approx 0.01$). A fair coin is maximally suspenseful: $0.5 \times 0.5 = 0.25$. Variance is largest exactly when you are most unsure. This one curve is worth remembering — it returns in election polling, A/B testing, and sample-size planning in Session 7.

Worked example: the basketball shot (slide example)

A player scores with probability $p = 0.6$. What is the probability he *misses*? Misses are the “0” outcome, so $P(X = 0) = 1 - p = 0.4$. Trivial on purpose — the point is the labelling discipline: *you* choose what counts as success, then everything else follows mechanically.

Where you meet Bernoulli in real life: a single product passing or failing a quality check, one survey respondent answering yes or no, one loan defaulting or not, one visitor clicking an ad or not. Alone it is almost too simple to be useful — its real job is to be the **Lego brick** from which the next two distributions are built.

3 Binomial: Bernoulli on repeat

The intuition

Take one Bernoulli trial. Now run it n times independently and *count the successes*. That count, X , follows the Binomial distribution. “How many of 10 customers ask for water?” “How many heads in 10 tosses?” “How many of 5 guessed MCQs are right?” — all the same mould: $X \sim \text{Bin}(n, p)$.

The four conditions (the BINS check)

A situation is binomial only if all four hold:

- B Binary:** each trial ends in exactly one of two outcomes (success/failure);
- I Independent:** one trial’s result does not influence another’s;
- N Number fixed:** the number of trials n is decided in advance;
- S Same p :** the success probability is identical on every trial.

Watch out

The two slide examples where binomial *fails* are worth internalising, because exams love them.
(a) “Number of people catching Covid in a room of 30.” Infection spreads: if one person is infected, their neighbour’s chance rises. Trials are *not independent* — the I fails. Not binomial.
(b) “2 of 20 projectors are defective; we pick 3 without replacement.” After you remove a defective projector, the proportion of defectives among the rest changes — p does not stay constant, and the trials are not independent. Not binomial. (This situation has its own mould, the hypergeometric.) The give-away phrase is *sampling without replacement from a small batch*.

Reading the formula like a sentence

$$P(X = x) = \underbrace{\binom{n}{x}}_{\text{how many ways}} \underbrace{p^x}_{x \text{ successes}} \underbrace{q^{n-x}}_{n-x \text{ failures}}$$

Why these three factors? Suppose you want exactly 6 of 10 customers to ask for water. One *specific* arrangement — say the first six ask, the last four do not — has probability $p^6 q^4$, just multiplying independent events. But there are many arrangements that all give “exactly 6”: the askers could be any 6 seats out of 10. The binomial coefficient $\binom{10}{6} = 210$ counts those arrangements. Ways $\times$ probability of each way = total probability.

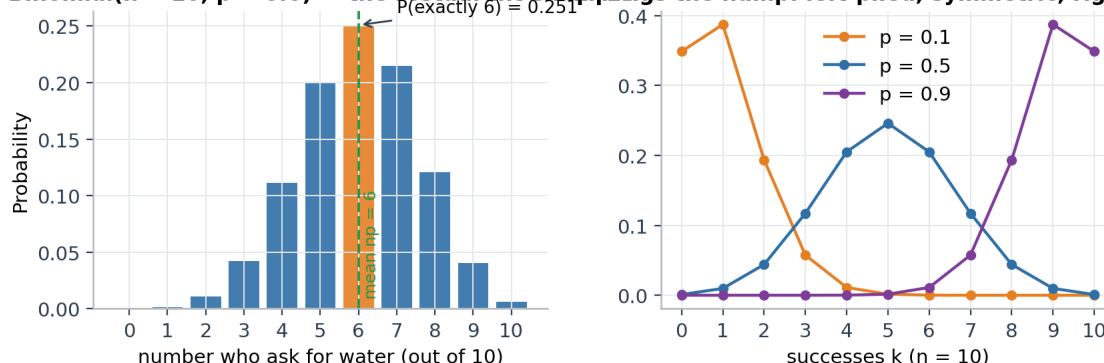
The mean and variance come almost free once you see X as a sum of n little Bernoulli bricks $Y_1 + \dots + Y_n$:

$$E[X] = np, \quad \text{Var}(X) = npq.$$

Each brick contributes p of expected success and pq of variance; n bricks contribute n times as much.

The mean np is exactly what common sense says: 10 customers at 60% each $\Rightarrow$ expect 6.

Binomial($n = 10, p = 0.6$) — the restaurant example has the hump: left-piled, symmetric, right-piled



The phrase translator

Most binomial marks are lost not on the formula but on translating English into events. Keep this table at hand:

The question says	You compute
"exactly 6"	$P(6)$
"at most 2"	$P(0) + P(1) + P(2)$
"fewer than 9"	$P(0) + \dots + P(8) = 1 - P(9) - P(10)$
"at least 3"	$1 - [P(0) + P(1) + P(2)]$ — the complement shortcut
"no one"	$P(0) = q^n$
"at least one"	$1 - P(0) = 1 - q^n$ — the single most used trick

The complement shortcut matters because adding eight probabilities is slow and error-prone, while subtracting two from 1 is fast. Whenever the list of outcomes you want is longer than the list you do not want, flip to the complement.

Worked example: the restaurant (slide Example 1, all five parts)

3 of every 5 customers ask for water; 10 customers are sampled. So $n = 10, p = 0.6, q = 0.4$.

(i) **Exactly 6:** $P(6) = \binom{10}{6}(0.6)^6(0.4)^4 = 210 \times 0.046656 \times 0.0256 = \mathbf{0.251}$.

(ii) **Fewer than 9:** the complement is "9 or 10", so $P(X < 9) = 1 - P(9) - P(10) = 1 - 0.0403 - 0.0060 = \mathbf{0.954}$.

(iii) **No one:** $P(0) = (0.4)^{10} = \mathbf{0.0001}$. With $p = 0.6$ per head, all ten declining is a one-in-ten-thousand event.

(iv) **At most 2:** $P(0) + P(1) + P(2) = 0.0001 + 0.0016 + 0.0106 = \mathbf{0.0123}$.

(v) **At least 3:** complement of (iv): $1 - 0.0123 = \mathbf{0.9877}$.

Notice how (v) recycles (iv) — structuring your work pays.

Worked example: three or fewer heads in 10 tosses (slide Example 2)

$n = 10, p = q = 0.5$. Because $p = q$, every term shares the factor $(0.5)^{10} = 1/1024$:

$$P(X \leq 3) = \frac{\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3}}{1024} = \frac{1 + 10 + 45 + 120}{1024} = \frac{176}{1024} = \mathbf{0.172}.$$

For fair coins, binomial probability is pure counting: favourable arrangements over 2^n .

 Worked example: dice product equal to 6 (slide Example 3)

A pair of dice is rolled 5 times; “success” = the product of the two faces is 6. First find p : the products equal to 6 come from $(1, 6), (6, 1), (2, 3), (3, 2)$ — 4 of the 36 equally likely pairs, so $p = 4/36 = 1/9, q = 8/9$. Now “at least 4 successes in 5 tries”:

$$P(X \geq 4) = P(4) + P(5) = \binom{5}{4} \left(\frac{1}{9}\right)^4 \left(\frac{8}{9}\right) + \left(\frac{1}{9}\right)^5 = \frac{40 + 1}{9^5} = \frac{41}{59049} \approx \mathbf{0.00069}.$$

Two lessons: (1) often the real work is computing p from a smaller probability puzzle; (2) rare things repeated rarely stay rare — a result we formalise with Poisson next.

 Worked example: guessing an MCQ exam (slide Example 4)

5 questions, 4 choices each, pure guessing: $n = 5, p = 1/4, q = 3/4$.

(a) **Exactly 3 correct:** $P(3) = \binom{5}{3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2 = 10 \cdot \frac{1}{64} \cdot \frac{9}{16} = \frac{45}{512} = \mathbf{0.088}$.

(b) **At most 3:** complement is “4 or 5 correct”: $1 - \left[\binom{5}{4} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right) + \left(\frac{1}{4}\right)^5\right] = 1 - \frac{15+1}{1024} = \frac{63}{64} = \mathbf{0.984}$.

(c) **At least one:** $1 - q^n = 1 - \left(\frac{3}{4}\right)^5 = 1 - \frac{243}{1024} = \mathbf{0.763}$.

(d) Mean $np = 1.25$ correct answers; variance $npq = 15/16 = 0.94$. A guesser passes a 3-out-of-5 bar with probability under 10% — the casino always wins.

4 Poisson: counting rare events in a window

 The intuition

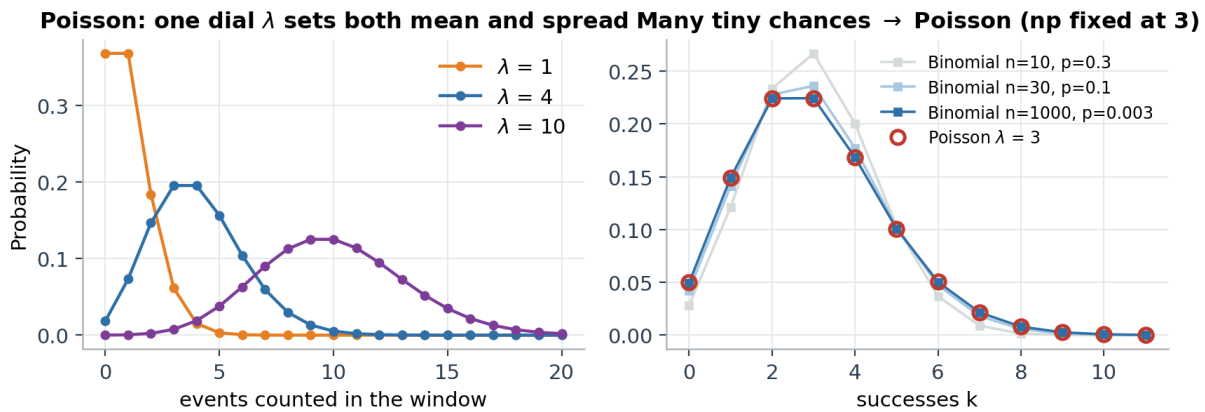
Binomial counts successes in n labelled trials. But many real questions have no visible trials at all: “How many calls hit the switchboard in 9 minutes?” “How many typos on a page?” “How many potholes per kilometre?” There is no n — just a *window* of time or space and events landing in it. The Poisson distribution covers exactly this: **counts of rare, independent events in a fixed window**, governed by one single number λ , the average count per window.

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots \quad E[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

★ **Everyday picture**

Raisins in dough. A baker stirs 300 raisins into dough and cuts 100 buns. Each raisin had a tiny chance $(1/300)$ of landing in *your* bun, and there were many raisins acting independently. Count the raisins in your bun: Poisson with $\lambda = 3$. Calls hitting a switchboard are raisins landing in a one-minute bun. Typos are raisins in a page-sized bun. Whenever *many independent opportunities each with tiny probability* scatter across a window, Poisson appears.

Why is the variance *equal* to the mean? Because a Poisson count is secretly a binomial with a huge n and tiny p : $\text{Var} = npq = \lambda q$, and with $p \approx 0$ we have $q \approx 1$, so $\text{Var} \approx \lambda$. A useful diagnostic in practice: if your count data has variance far above its mean, plain Poisson is the wrong mould.



✓ Key takeaway

λ **must match the window**. The rate you are given and the window you are asked about rarely agree — rescale first, formula second:

$$\lambda = \text{rate} \times \text{window size}.$$

Calls at 2 per 3 minutes, asked about 9 minutes $\Rightarrow \lambda = \frac{2}{3} \times 9 = 6$. Defects at 5 per 10 sq ft, asked about 15 sq ft $\Rightarrow \lambda = 7.5$. Most Poisson errors in exams are window errors, not formula errors.

📎 Worked example: the switchboard (slide Example 5)

Calls arrive at 2 per 3 minutes. Probability of 5 or more calls in 9 minutes?

Step 1 — *rescale*: $\lambda = (2/3) \times 9 = 6$ calls expected in the window.

Step 2 — *complement*: “5 or more” has infinitely many terms; “4 or fewer” has five:

$$P(X \geq 5) = 1 - e^{-6} \left(1 + 6 + \frac{6^2}{2!} + \frac{6^3}{3!} + \frac{6^4}{4!} \right) = 1 - e^{-6} (1 + 6 + 18 + 36 + 54) = 1 - 0.2851 = \mathbf{0.715}.$$

📎 Worked example: defective units (slide Example 6)

3% of units are defective; in a sample of 200, find $P(\text{fewer than 2 defective})$.

This is really binomial ($n = 200, p = 0.03$) but n is large and p small, so Poisson with $\lambda = np = 6$ is an excellent stand-in:

$$P(X < 2) = P(0) + P(1) = e^{-6} + 6e^{-6} = 7e^{-6} = \mathbf{0.0173}.$$

Poisson as the binomial's limit

The slides state the official rule: when $n \rightarrow \infty$ and $p \rightarrow 0$ with $np = \lambda$ fixed, the binomial converges to Poisson; the approximation is already good when $n \geq 100$ and $np \leq 10$. The right panel of the figure above shows the convergence happening: by $n = 1000$ the binomial dots sit inside the Poisson circles.

📎 Worked example: defective bindings, both ways (slide Example 7)

5% of books have defective bindings. Probability exactly 2 of 100 books are defective?

Exact binomial: $P(2) = \binom{100}{2} (0.05)^2 (0.95)^{98} = 4950 \times 0.0025 \times 0.00655 = \mathbf{0.081}$.

Poisson stand-in: $\lambda = np = 5$, so $P(2) = e^{-5} 5^2 / 2! = 12.5 e^{-5} = \mathbf{0.084}$.

The two answers differ by 0.003 — the price of replacing a hundred-term computation with one line. That trade is the whole point of approximations.

 Worked example: the consultant's switchboard (slide Example 8)

Calls arrive at 0.6 per minute.

(a) **At least 1 call in a minute:** $\lambda = 0.6$, and $P(X \geq 1) = 1 - P(0) = 1 - e^{-0.6} = \mathbf{0.451}$.

(b) **At least 3 calls in 4 minutes:** rescale to $\lambda = 0.6 \times 4 = 2.4$, then

$$P(X \geq 3) = 1 - e^{-2.4} \left(1 + 2.4 + \frac{2.4^2}{2} \right) = 1 - e^{-2.4}(6.28) = \mathbf{0.430}.$$

 Worked example: the cafeteria (slide Example 12)

Customers arrive at 0.3 per minute.

(a) **Exactly 2 in 10 minutes:** $\lambda = 3$; $P(2) = e^{-3} 3^2 / 2! = 4.5 e^{-3} = \mathbf{0.224}$.

(b) **2 or more in 10 minutes:** $1 - e^{-3}(1 + 3) = 1 - 0.199 = \mathbf{0.801}$.

(c) **Exactly one in each of two back-to-back 5-minute spans:** each span has $\lambda = 1.5$ and $P(1) = 1.5 e^{-1.5} = 0.335$; disjoint windows are independent, so multiply: $0.335^2 = \mathbf{0.112}$. Independence across non-overlapping windows is a defining Poisson property — it is what lets us multiply.

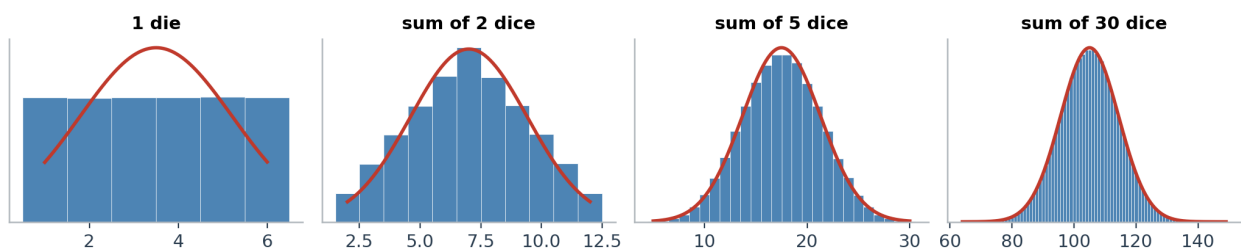
Where Poisson runs the world: arrivals at queues (people at airports, planes at runways, read/write requests at servers, loan applications at banks) and defects scattered over material (flaws per 1,000 ft of wire, blemishes per square foot, errors per page). Any time you hear “per hour”, “per page”, “per km” — think Poisson.

5 Normal: the shape that keeps showing up

 The intuition

Why does one bell-shaped curve describe heights, measurement errors, exam scores, daily returns, and filling weights? Because each of those quantities is the *sum of many small independent influences* — genes plus nutrition plus chance for height; hundreds of tiny hand tremors for a measurement. When many little nudges add up, their total always piles into the same bell, regardless of what the individual nudges look like. (Session 7 turns this observation into a theorem — the Central Limit Theorem.)

Add up enough independent random things and a bell appears



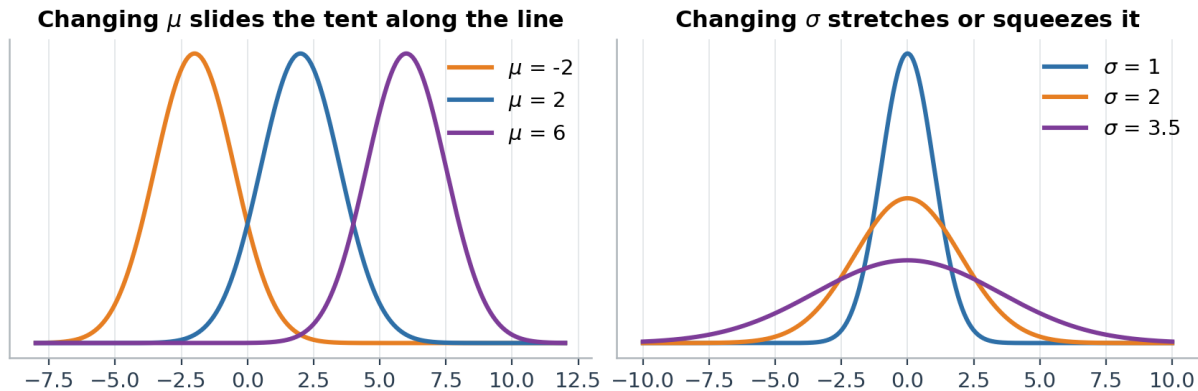
One die is flat — every face equally likely. The sum of just two dice already forms a triangle (there are more ways to make 7 than 2). By thirty dice the histogram is indistinguishable from the red normal curve. *Nothing about a die is bell-shaped; the bell lives in the act of adding.*

A continuous random variable X is normal with mean μ and variance σ^2 , written $X \sim N(\mu, \sigma^2)$, if

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty.$$

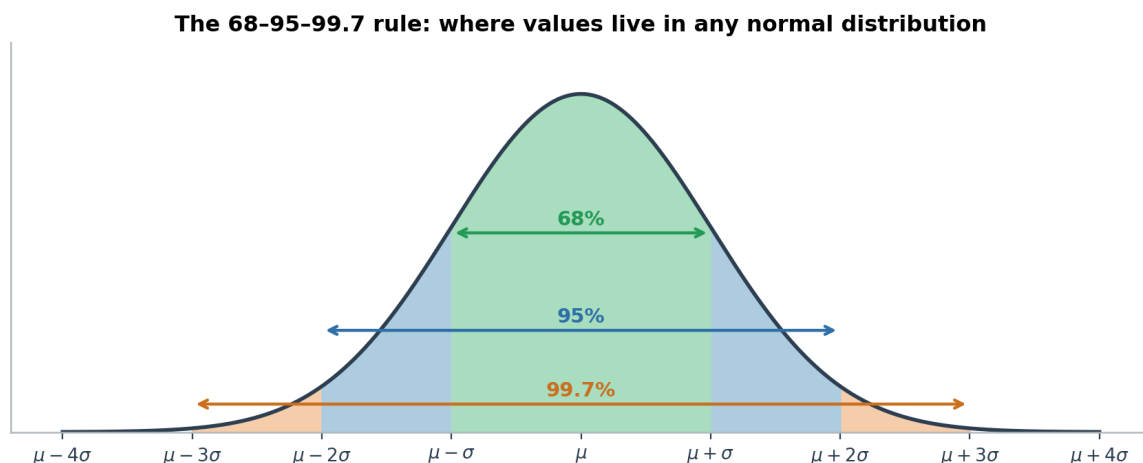
Do not let the formula intimidate you — read it as a recipe. The core is $e^{-\frac{1}{2}z^2}$ where z counts how many σ -steps x is from μ : at the mean the exponent is 0 and the curve peaks; move away and the negative quadratic exponent crushes the height, faster and faster. The constant $1/(\sigma\sqrt{2\pi})$ merely rescales so the total area is exactly 1, as every probability distribution requires.

Only two knobs. μ says *where* the tent is pitched; σ says *how wide* it sags:



Properties worth reciting: the curve is symmetric about μ , so mean = median = mode; total area is 1; the curve approaches but never touches the axis (any value is *possible*, extreme values are just absurdly unlikely). And for continuous variables, probability *is area*: $P(a < X < b)$ is the area under the curve between a and b ; a single exact point has area zero, which is why $P(X < a)$ and $P(X \leq a)$ are the same for the normal.

The 68–95–99.7 rule: a free probability calculator



About 68% of values fall within 1σ of the mean, 95% within 2σ , 99.7% within 3σ — *for every normal distribution, whatever μ and σ are*. If adult heights average 170 cm with $\sigma = 7$: two-thirds of people are 163–177 cm, 19 in 20 are 156–184 cm, and meeting someone outside 149–191 cm is a 3-in-1000 event. The rule also powers instant sanity checks: a data point “4 sigmas out” should make you suspect a typo before you believe a miracle.

6 Z-scores: one table to rule them all

The intuition

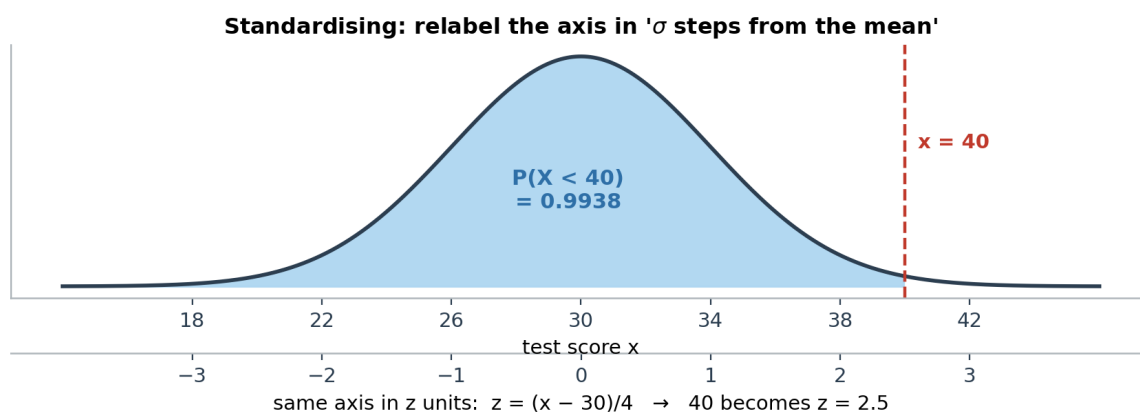
There are infinitely many normal curves — one per (μ, σ) pair. We cannot print a probability table for each. The fix: convert every normal question into the *same currency*,

$$Z = \frac{X - \mu}{\sigma},$$

which asks: **how many standard deviations is x above the mean?** Subtracting μ re-centres the curve at 0; dividing by σ makes one step equal one standard deviation. The result is the *standard normal* $Z \sim N(0, 1)$, and a single table of its areas, $F(z) = P(Z \leq z)$, answers every normal question ever asked.

★ Everyday picture

A z-score is a currency conversion. Scoring 75 in a test with mean 60, $\sigma = 10$ and scoring 700 in another with mean 600, $\sigma = 100$ sound incomparable — until you convert both to the common currency: $z = 1.5$ and $z = 1.0$. Now you know the first performance was actually stronger. The slides prove the conversion works in general: if $X \sim N(\mu, \sigma^2)$ then $E(Z) = 0$, $\text{Var}(Z) = 1$, and more generally any linear transform $aX + b$ of a normal stays normal with mean $a\mu + b$ and variance $a^2\sigma^2$.



Working ritual (use it every single time): (1) sketch a bell, mark μ ; (2) mark your value(s) x and shade the area the question asks for; (3) convert to z ; (4) read $F(z)$ from the table; (5) assemble using two facts only:

$$P(Z > z) = 1 - F(z), \quad F(-z) = 1 - F(z) \text{ (mirror symmetry).}$$

Worked example: reading the table four ways (slide Example 9)

Z is standard normal. **(a)** $P(Z < 1.75) = F(1.75) = 0.9599$ — direct lookup. **(b)** $P(Z < -1.25) = 1 - F(1.25) = 1 - 0.8944 = 0.1056$ — mirror. **(c)** $P(Z > 2.06) = 1 - F(2.06) = 0.0197$ — complement. **(d)** $P(Z > -1.82) = F(1.82) = 0.9656$ — mirror then complement, or just note that “above a negative value” is the big region.

 Worked example: test scores (slide Examples 10–12, condensed)

Scores are $N(30, 4^2)$.

Below 40: $z = (40 - 30)/4 = 2.5$, so $P = F(2.5) = 0.9938$.

Above 38: $z = 2$, so $P = 1 - F(2) = 1 - 0.9773 = 0.0227$. (The empirical rule predicted this: above $+2\sigma$ lives about 2.5%.)

Between 70 and 80 when scores are $N(75, 4^2)$: the bounds sit symmetrically at $z = \mp 1.25$, so $P = F(1.25) - F(-1.25) = 0.8944 - 0.1056 = 0.7888$.

 Worked example: two-sided regions (slide Example 13)

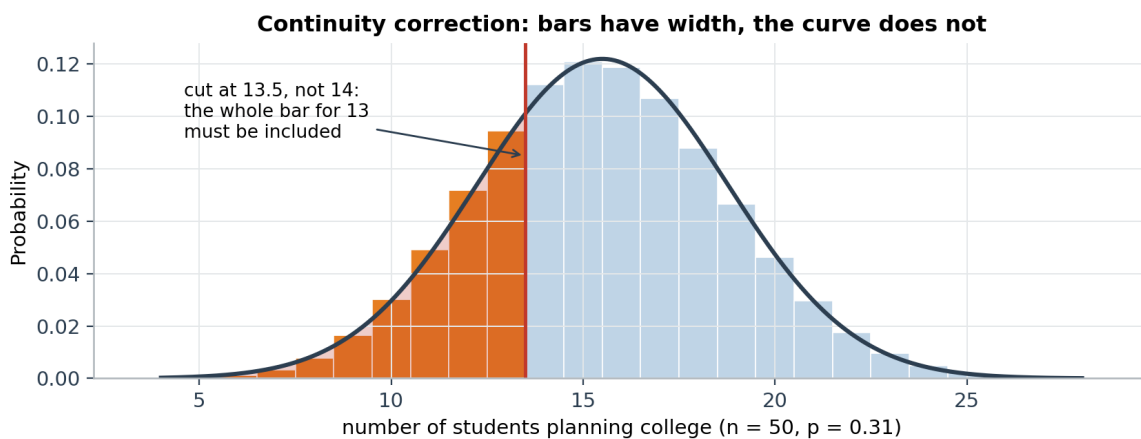
$X \sim N(30, 5^2)$. **(i)** $P(26 \leq X \leq 40)$: z -bounds $(26 - 30)/5 = -0.8$ and $(40 - 30)/5 = 2$; $F(2) - F(-0.8) = 0.9773 - 0.2119 = 0.765$. **(ii)** $P(X > 45)$: $z = 3$; $1 - F(3) = 1 - 0.9986 = 0.0014$ — the 99.7 rule again.

7 When Binomial wears the bell: normal approximation

Computing $P(X < 14)$ for $\text{Bin}(50, 0.31)$ honestly means summing fourteen binomial terms. But look at the bars in the figure below — they already trace a bell. When n is large enough, we simply *ride the bell instead of adding the bars*:

$$X \approx N(\mu = np, \sigma = \sqrt{npq}) \quad \text{valid when } np \geq 15 \text{ and } nq \geq 15.$$

The two conditions guarantee the hump sits far enough from 0 and n that the bell is not cut off at either end. (Compare: 25 people, 36% dog owners gives $np = 9 < 15$ — refuse to approximate, as the slide example does.)



 Key takeaway

Continuity correction — the half-step of honesty. Binomial bars are blocks of width 1 centred on integers; the bar for “13” occupies 12.5 to 13.5. A continuous curve must include the *whole block* of every integer it claims to count:

$$P(X \leq k) \rightarrow P(X \leq k + 0.5), \quad P(X < k) \rightarrow P(X \leq k - 0.5),$$

$$P(X \geq k) \rightarrow P(X \geq k - 0.5), \quad P(X > k) \rightarrow P(X \geq k + 0.5).$$

Quick self-check: ask “which integers do I want?” and stretch half a unit beyond the outermost one you keep.

 Worked example: college plans (slide Example 15)

31% of seniors plan to attend college; 50 students sampled; find $P(X < 14)$.

Check: $np = 15.5 \geq 15$, $nq = 34.5 \geq 15$ — bell approved. $\mu = 15.5$, $\sigma = \sqrt{50 \times 0.31 \times 0.69} = 3.27$.

Correct: “fewer than 14” keeps the integers up to 13, so cut at 13.5.

Convert: $z = (13.5 - 15.5)/3.27 = -0.61$, giving $P = F(-0.61) = \mathbf{0.2709}$.

 Worked example: cloud seeding (slide Example 14)

62% of seeded clouds show spectacular growth; of 40 seeded clouds, find $P(X \leq 20)$.

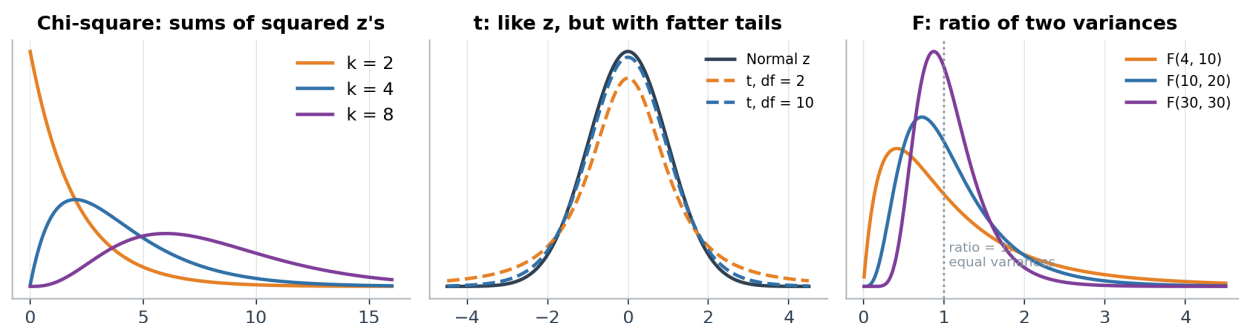
$np = 24.8$, $nq = 15.2$ — both at least 15. $\mu = 24.8$, $\sigma = \sqrt{9.424} = 3.07$. With the correction, $P(X \leq 20) \rightarrow P(X \leq 20.5)$:

$$z = \frac{20.5 - 24.8}{3.07} = -1.40, \quad P = F(-1.40) = \mathbf{0.0808}.$$

Note: the slide prints 0.808 — a slipped decimal. Sanity check it yourself: 20 successes is 1.4σ below the mean, so the probability must be well under one half. Cultivate exactly this reflex.

8 First hello to the inference trio: χ^2 , t , F

These three distributions are not new shapes of data — they are shapes of *statistics we compute from samples*. All three are manufactured from standard normal ingredients, and each answers one recurring question in inference. Today is just the introduction; they take centre stage in the coming sessions.



Chi-square: total squared surprise

Take k independent standard normals and add their squares:

$$Y = z_1^2 + z_2^2 + \cdots + z_k^2 \sim \chi_k^2.$$

Each z_i^2 is one “squared surprise”; Y is the total. Squares cannot be negative, so the curve lives entirely to the right of zero and is skewed — occasionally one large z blows the total up. Its one parameter k is called the **degrees of freedom** (how many independent squares went in). Because sample variance is itself a sum of squared deviations, χ^2 is the natural home of *variance estimation* and *goodness-of-fit* tests (“do my observed counts deviate from the expected ones by more than chance?”).

Student's t: the bell with a humility tax

The z-score needs the true σ . In real life you almost never have σ — you estimate it with the sample standard deviation S , computed with the $n - 1$ divisor. Replacing a known constant by a noisy estimate injects extra uncertainty, and the t-distribution prices it in:

$$t = \frac{z}{\sqrt{\chi_k^2/k}} \quad \text{and concretely} \quad t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

The shape is a bell, but with **heavier tails** — it admits “my estimate of σ might itself be off, so extreme outcomes deserve more probability”. Small sample $\Rightarrow$ noisy $S \Rightarrow$ fat tails; as n grows, S locks onto σ and the t-curve melts into the normal (visible in the middle panel: $df = 10$ is already close).

F: a tug-of-war between two variances

Take two independent chi-squares, divide each by its degrees of freedom, then take the ratio:

$$F = \frac{\chi_{v_1}^2/v_1}{\chi_{v_2}^2/v_2}, \quad \text{in practice} \quad F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1, n_2-1}.$$

This is the tool for “*which population is more variable?*” — two machines, two markets, two labs. If the variances are truly equal the ratio should hover near 1; a ratio far from 1 is evidence of unequal variability. Note the order of the degrees of freedom matters ($F_{4,10} \neq F_{10,4}$), and the curve again lives only on the positive axis. F is the engine behind ANOVA, coming later in the course.

✓ Key takeaway

A one-line map to remember: **z** when σ is known; **t** when σ is estimated; χ^2 for one variance and goodness of fit; **F** for comparing two variances.

9 Practice problems from the slides, decoded

P1 — machines breaking down (Poisson approximation)

8% of machines break down; 25 machines. $\lambda = np = 2$. (a) $P(5) = e^{-2}2^5/5! = e^{-2}(32/120) = \mathbf{0.036}$. (b) $P(X \geq 4) = 1 - e^{-2}(1 + 2 + 2 + \frac{4}{3}) = 1 - 0.857 = \mathbf{0.143}$. (c) $P(3 \leq X \leq 8) = e^{-2} \sum_{k=3}^8 \frac{2^k}{k!} = \mathbf{0.323}$.

P2 — trucks at a depot

$\lambda = 12$ per day; “fewer than nine” means $P(X \leq 8) = e^{-12} \sum_{k=0}^8 \frac{12^k}{k!} = \mathbf{0.155}$. Tedious by hand but mechanical — and a perfect candidate for the complement *not* being shorter, so you just add.

P3 — sheet metal defects (window rescale!)

5 defects per 10 sq ft; sheet of 15 sq ft $\Rightarrow \lambda = 7.5$. $P(X \geq 6) = 1 - e^{-7.5} \sum_{k=0}^5 \frac{7.5^k}{k!} = 1 - 0.2414 = \mathbf{0.759}$.

P4 — standard normal warm-ups

(i) $P(Z > 1.09) = 1 - 0.8621 = \mathbf{0.1379}$; (ii) $P(Z < -1.65) = \mathbf{0.0495}$; (iii) $P(-1.00 < Z < 1.96) = 0.9750 - 0.1587 = \mathbf{0.8163}$; (iv) $P(1.25 < Z < 2.75) = 0.9970 - 0.8944 = \mathbf{0.1026}$.

P5 — instrument lifetimes

Life $\sim N(12, 2^2)$ months. (a) $P(X < 7)$: $z = -2.5 \Rightarrow \mathbf{0.0062}$. (b) $P(7 < X < 12) = F(0) - F(-2.5) = 0.5 - 0.0062 = \mathbf{0.4938}$.

P6 — the crashing computer (a lesson in windows)

“Crashes on average once every 4 months” over a 4-month window gives $\lambda = \text{rate} \times \text{window} = \frac{1}{4} \times 4 = 1$. Then $P(0) = e^{-1} = \mathbf{0.368}$, $P(1) = e^{-1} = \mathbf{0.368}$, $P(2) = e^{-1}/2 = \mathbf{0.184}$, $P(3) = e^{-1}/6 = \mathbf{0.061}$. *Heads-up*: the slide’s hint plugs $\lambda = 4$, which would describe *four* expected crashes per window — a different machine! Whenever your λ disagrees with a hint, re-read the window. Trust rate $\times$ window.

P7–P9 — binomial quickies

P7: $n = 30$, $np = 20 \Rightarrow p = \frac{2}{3}$, so $\text{Var} = npq = 20 \times \frac{1}{3} = \mathbf{6.67}$, $\text{SD} = \mathbf{2.58}$. **P8**: want $P(\text{at least one hit}) > \frac{2}{3}$ with $p = 0.2$: $1 - 0.8^n > \frac{2}{3} \Leftrightarrow 0.8^n < \frac{1}{3}$; since $0.8^4 = 0.41$ and $0.8^5 = 0.33$, the boy must fire $n = 5$ times. **P9**: pink candies: $P(4) = \binom{10}{4}(0.15)^4(0.85)^6 = \mathbf{0.040}$.

P10 — reverse-engineering a binomial

Mean 6, variance 4: divide! $\frac{npq}{np} = q = \frac{4}{6} = \frac{2}{3}$, so $p = \frac{1}{3}$ and $n = 6/p = 18$. The distribution is $\text{Bin}(18, \frac{1}{3})$. Dividing variance by mean is the standard unlock for these.

P11 — defective parts across 1000 samples

Mean defectives in samples of 20 is 2 $\Rightarrow p = 0.1$. $P(X \geq 3) = 1 - [(0.9)^{20} + 20(0.1)(0.9)^{19} + 190(0.01)(0.9)^{18}] = 1 - 0.677 = 0.323$. Out of 1000 samples, expect $0.323 \times 1000 = \mathbf{323}$ samples with at least 3 defectives — note the elegant final move: a probability becomes a forecast count by multiplying by the number of repetitions.

P13 — valve flow coefficients

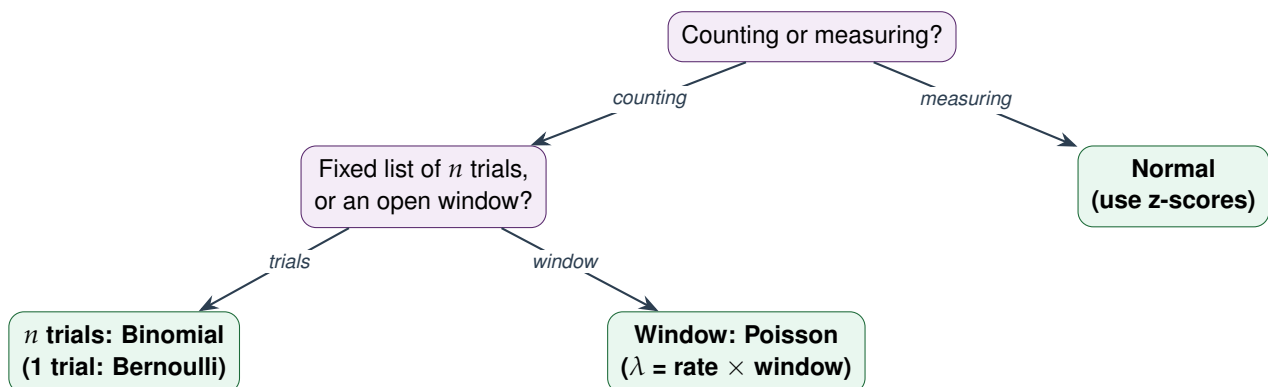
$X \sim N(496, 25^2)$. (a) $P(X > 450)$: $z = -1.84$; $1 - F(-1.84) = F(1.84) = \mathbf{0.967}$. (b) $P(445.5 < X < 522)$: z -bounds -2.02 and 1.04 ; $0.8508 - 0.0217 = \mathbf{0.829}$.

10 The one-page cheat sheet

Mould	Use when...	PMF / PDF	Mean	Variance
Bernoulli(p)	one trial, two outcomes	$p^x q^{1-x}$	p	pq
Binomial(n, p)	count successes in n fixed independent trials, constant p	$\binom{n}{x} p^x q^{n-x}$	np	npq
Poisson(λ)	count events in a time/space window; rare events, many chances	$\frac{e^{-\lambda} \lambda^x}{x!}$	λ	λ
Normal(μ, σ^2)	measurements; sums of many small effects	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	μ	σ^2

Bridges between moulds: Binomial $\rightarrow$ Poisson when $n \geq 100$, $np \leq 10$ (use $\lambda = np$); Binomial $\rightarrow$ Normal when $np \geq 15$ and $nq \geq 15$ (use $\mu = np$, $\sigma = \sqrt{npq}$, and never forget the ± 0.5 continuity correction).

The inference trio: χ_k^2 = sum of k squared z 's (variances, goodness of fit); $t_{n-1} = z$ with estimated σ (heavier tails, small samples); F_{v_1, v_2} = ratio of scaled χ^2 's (comparing two variances).



✓ Key takeaway

The exam-day routine: **(1)** name the mould (check BINS for binomial; check window for Poisson; check $np, nq \geq 15$ before riding the bell); **(2)** write down its parameters from the words of the problem; **(3)** translate the English phrase into an event, using complements to shorten work; **(4)** compute, and **(5)** sanity-check the answer against the mean and the 68–95–99.7 rule.